

Ideation Phase

Define the Problem Statements

Date	05 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	2 Marks

Customer Problem Statement Template:

Defining the Problem Statement is a crucial step in understanding the challenges faced by users in this project. The main objective is to analyze EV charging behaviour, battery range, energy consumption, and infrastructure availability using Tableau for effective visualization and decision-making.

The core problem is that EV owners and fleet operators lack clear, interactive insights into charging patterns and range performance because data is often scattered across multiple sources and difficult to analyze. This project addresses the issue by creating Tableau dashboards that visualize charge behaviour, range efficiency, and charging infrastructure, helping users make informed

I am	Describe customer with 3-4 key characteristics – who are they?	EV owner with specific driving habits, and their attributes here
I'm trying to	Get their outcome or "job" the care about – what are they trying to achieve?	Optimize driving and minimize charging time to achieve here
but	Describe what problems or barriers stand in the way – what bothers them most?	Anxious about unexpected battery depletion, but get in the way here
because	Enter the "root cause" of why the problem or barrier exists – what needs to be solved?	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – how does it impact them emotionally?	Describe the emotions the result from experiencing the problems or barriers

Problem Statement 1: Anxiety-Prone Urban EV Commuter

I am	I'm trying to	But	Because	Which makes me feel
an urban commuter who wants to maintain- wants in managing a balance if EV commute.	predict my daily travel range and plan charging to untion my healthier healthier choices.	range estimates can be inconsistent, and I can't have a way to analyze daily habits and nutrition.	there is no single tool that interactive dashboard that provides visual insights into my food consumption and nutritional trends.	anxious about driving in cold weather.. I am unable to informed informed decisions and maintain a balanced lifestyle.

Problem statement 2:Fleet Operator/EV Manager

I am	I'm trying to	But	Because	Which makes me feel
 a fleet operator who manages multiple EVs and charging operations.	 analyze charging patterns, monitor mng performance, and optimize energy usage and charging schedules.	 data is scattered across systems and lacks clear visualization of usage, efficiency, and infrastructure.	 there is no unified dashboard that integrates charging sessions, range analytics, and charger availability.	 overwhelmed and inefficient, making it hard to optimize fleet operations and reduce costs.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1 EV Owner (Daily Commuter)	I am an EV owner who commutes daily and wants to manage charging efficiently.	I'm trying to plan my trips confidently, monitor my battery range, and find the best charging options.	But my range is unpredictable, charging data is scattered, and I can't easily compare charging stations.	Because there is no centralized, interactive tool that combines range, charging behavior, and infrastructure data.	Which makes me feel anxious and inconvenienced, worrying about running out of charge and wasting time searching for chargers.
PS-2 Fleet Operator / EV Manager	I am a fleet operator who manages multiple EVs and charging operations.	I'm trying to analyze charging patterns, monitor range performance, and optimize energy usage and charging schedules.	But data is scattered across systems and lacks clear visualization of usage, efficiency, and infrastructure.	Because there is no unified dashboard that integrates charging sessions, range analytics, and charger availability.	Which makes me feel overwhelmed and inefficient, making it hard to optimize fleet operations and reduce costs.